YUAN-JYUN LEE

Contact Information

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Research Interests

Functional analysis, Quantum information theory, Matrix analysis, Operator theory, Free probability

Education

National Yang Ming Chiao Tung University (NYCU), Taiwan *Fall '22 - Fall '26*
B.S. in Department of Applied Mathematics
Cumulative GPA: 4.02/4.30 | 4.00 Scale Equivalent: 3.89
Ranking: 2/55 (Top 3.64%)

National Yang Ming Chiao Tung University (NYCU), Taiwan *Fall '23 - Fall '26*
B.S. in Department of Electronics and Electrical Engineering

Some Graduate Coursework
Functional Analysis, Operator Theory II, Real Analysis I & II (Measure Theory), Knot theory, Quantum Mechanics, Algebraic Graph Theory, Stochastic Process, Fluid Dynamics

Research Experience

Undergraduate Summer Research Program, NCTS, Taipei, Taiwan *summer 2026*
Harmonic Analysis and Combinatorics in Free Probability Theory
Advisor: Hao-Wei, Huang, Pei-Lun Tseng, Jiun-Chau Wang

- Study the fundamentals in free probability from analysis and combinatorics viewpoints.
- Compute several applications in Quantum information theory about characterizing entanglement.
- Primary reference: Nica and Speicher, *Lectures on the Combinatorics of Free Probability* (2006)

Reading group on open quantum system, Hsinchu, Taiwan *2025*
Study in quantum markov semigroup
Advisor: Cheng-Fang Su

- We study the basic of open quantum system, mainly stduying the markovian quantum evolution.
- Construct explicitly models of euqation in lower dimensional cases.

Reading group on ergodic theory, Hsinchu, Taiwan *Summer '25*
Studied foundational results in ergodic theory
Advisor: Xiang Fang

- We have studied basic knowledge in ergodic theory, starting from measure preseving maps, ergodicity, to weakly mixing…
- Several examples are computed.
- Primary reference: Walters, *An Introduction to Ergodic Theory* (2000).

Undergraduate Summer Research Program, TWSIAM, Hsinchu, Taiwan *Summer '25*
Classical and Modern perspectives on numerical range
Advisor:Kuo Zhong Wang

- We have learned some results in numerical range of operators mainly on finite dimentional.
- Important results such as Hausdorff–Toeplitz theorem have been reviewed.
- Studied some explicit examples in 2x2, 3x3 matrices.
- Primary reference: Pei Yuan Wu, *Numerical Ranges of Hilbert Space Operators*(2021).

Undergraduate Summer Research Program, TWSIAM, Hsinchu, Taiwan *Fall '24 - Spring '25*
A Study on Trace Inequalities Derived from Relative Entropy
Advisor: Cheng-Fang Su, Hao-Wei Huang

- Studied mathematical foundations of quantum information theory, focusing on entropy inequalities.
- Established trace inequality results under structural constraints on density matrices, contributing to ongoing research in the area.

Undergraduate Summer Research Program, TWSIAM, Hsinchu, Taiwan *Summer '23*
A research on finite time Kuramoto model
Advisor: Ming-Cheng Shiue
- Studied synchronization phenomena in the Kuramoto model.
- Have conducted some explicit numerical experiences in finite time variants.

Honors

College of Science Dean Award, Hsinchu, Taiwan *2024*
For ranking in the top 10% of the department in my first three years of undergraduate study.

Honor Student Distinction, Hsinchu, Taiwan *Spring '22, Fall '23, Spring '25*
Awarded for exceptional academic excellence (Top 5% of the department).

Mathematics Department Award, Hsinchu, Taiwan *'22, '23, '24*
Awarded by NYCU Applied Mathematics Department for outstanding performance

Excellence in Undergraduate Research, Hsinchu, Taiwan *2025*
Awarded for outstanding results in undergraduate research projects

Hsiao Cheng-Tang Memorial Scholarship, Hsinchu, Taiwan *2025*
Granted to the single most outstanding undergraduate student with high research potential

Zhu Shun-Yi Cooperative Scholarship, Hsinchu, Taiwan *2025*
Scholarship for academic excellence and contribution to research

Conference

Oral Presentation, Finite time convergence of Kuramoto model
2024 NTNU Fall Applied Mathematics Student Conference
Recipient of Best Presentation Award.

Teaching Experience

Calculus I, Teaching Assistant *Fall 2025*
Professor: Xiang Fang

Probability, Teaching Assistant *Fall 2025*
Professor: Xiang Fang

Fundamental Mathematics, Teaching Assistant *Fall 2025*
Professor: Kuo Zhong Wang

English

TOEFL(ibt): 5.5 (110) (CEFR: C1) *Test date: Sept. 2026*
Listening: 6 | Reading: 5
Speaking: 5 | Writing: 5

Skills

Programming: C++, Python, Matlab
Languages: Mandarin, English